

Danium Shahnawaz Syed

Applied Cybersecurity

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Summary

Passionate Security Engineer and Erasmus Mundus Scholar. Solid experience in DevOps, System Administration, and security research. Holding a double Master's degree in Cybersecurity and a CCNA certification.

Education

Kadir Has University & SRH Berlin University of Applied Sciences, Erasmus Mundus Double Master's in Applied Cybersecurity (CyberMACS) – Istanbul & Berlin Sept 2024 – Sept 2026

- Selected among the top 32 students from over 1,000 applications
- Received a fully funded scholarship given to the top applicants
- GPA: 3.48/4.00 (KHAS), 1.8 (SRH)

American International University-Bangladesh (AIUB), BSc in Electrical and Electronics Engineering – Dhaka, Bangladesh Sept 2019 – Sept 2023

- Received multiple awards and studied on a full scholarship since admission
- Graduated top of class with a CGPA of 3.98/4.00 (1.03 German scale)
- Received Best Project Award for Thesis/Capstone Project

Mount Royal Academy, IGCSE and A-Levels in Pearson Edexcel British Curriculum – Bangladesh June 2017 – June 2019

Experience

Cybersecurity Intern, Cyberforce – Istanbul, Turkey June 2025 – Sept 2025

- Developed internal tools for security testing of LLM/ML models from PyTorch, TensorFlow, Keras, Sklearn, and XGBoost, detecting insecure serialization formats inside Docker containers and malware-corrupted models to prevent model serialization attacks
- Conducted on-site and offline penetration testing of ML and LLM models; completed all levels of the Gandalf and Prompt Airlines CTFs
- Penetration tested various HackTheBox systems, gaining practical knowledge of web security

ICT and DevOps Intern, Grameenphone (Telenor) – Dhaka, Bangladesh Jan 2023 – June 2023

- Automated ICT workflows and built an internal tool to monitor monthly product revenues across 8 products, presenting them on a dashboard
- Worked on the system architecture formation of Grameenphone Alo, a proprietary IoT platform
- Extensively worked on hybrid-cloud architectures and Kubernetes, and on secure CI/CD deployment through Jenkins, Docker, and Terraform
- Conducted on-site surveys of Grameenphone datacenters

IT Educator, Dignitus International School – Narayanganj, Bangladesh July 2017 – July 2019

- Educated students from grade 8 to 10, with multiple A* achievements in taught subjects for board exams
- Maintained lesson plans and liaised with the international education board Edexcel

Publications

- Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains** Nov 2026
Danium S. Syed
(Submitted to SECITC (Springer LNCS) conference proceedings)
- Analyzing Political Sentiment on Twitter with Zero-Shot Large Language Models** Oct 2026
R. Ayoub, *Danium S. Syed*, E. O. Njoku
(Accepted in Proc. IACyC 2026)
- Explainable Botnet Detection Using Flow-Based Random Forest Classification and OSINT Validation** Apr 2026
A. Razan, E. O. Njoku, *Danium S. Syed*, N. Nachevski
(Web Proc. CIIT 2026)
- Reputation and Energy-Aware Dynamic Hybrid Consensus (READ-HC) Model for IIoT** July 2025
S. Mitra, V. S. Sable, *Danium S. Syed*
[10.31585/jbba-8-2-\(3\)2025](https://doi.org/10.31585/jbba-8-2-(3)2025) (The Journal of The British Blockchain Association)
- Design and Feasibility Analysis of a LoRa Based Communication System for Disaster Management** Aug 2024
Danium S. Syed, S. B. Hossain, A. Islam, T. A. I. Jim, M. Saniat, R. Zishan
[10.53799/ajse.v23i2.1055](https://doi.org/10.53799/ajse.v23i2.1055) (AIUB Journal of Science and Engineering (AJSE), 23, 135-144)

Projects

- Master's Thesis: Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains** Jan 2026 – present
- Conducted security analysis of containerized ML systems to measure prevalence and distribution of embedded model artifacts across Docker ecosystems
 - Identified security risks in ML supply chains by statically analyzing model files (PyTorch, TensorFlow, ONNX, Pickle) for unsafe serialization and code execution vulnerabilities
 - Built an automated pipeline to extract container filesystems and scan ML artifacts, generating empirical insights into real-world deployment risks
- Linux Home Server (HomeLab)** Ongoing
- Built from scratch and maintains a Debian-based home server for security research, home automation, and media consumption
 - Follows DevOps practices and maintains high uptime
 - Technology stack: WireGuard, Docker Compose, Proxmox
- Bachelor's Capstone: Long-Range Off-Grid Communication System Deployed with Amphibious Rovers** Jan 2023 – Sept 2023
- Built a mesh communication system using LoRa modulation for deployment in disaster scenarios via amphibious rovers
 - Provides text messaging and location-based communication services for affected users in disaster zones

Honors and Awards

- 5th Place (All Solves), CYBERUS Spring School CTF Competition — Lorient, France (2026)
- 2nd Place (All Solves), CyberMACS Winter School CTF Competition — Istanbul, Turkey (2025)
- Erasmus Mundus Full Scholarship, EMJM in Applied Cybersecurity — Turkey/Germany (2024)
- Dean's Best Project Award, AIUB — Dhaka, Bangladesh (2023)
- Dean's List Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Freedom Fighter's Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Edexcel High Achiever's Award, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)
- Featured in The Daily Star and The Daily Sun, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)

Certifications

- Building Resilience: Navigating Vulnerability, Risks, and Continuity in a Dynamic World — CyberMACS (2026)
- Cisco Certified Network Associate (CCNA), CSC014934967 — Cisco (2025)
- The Bits and Bytes of Computer Networking — Google/Cisco, Coursera (2020)

Skills

DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

System Administration: Linux, Networking, Virtualization, Containers, Git

Privacy & AI Security: Differential Privacy, Agentic AI, PyTorch, TensorFlow, vLLM/Ollama

Security Research: Vulnerability Analysis, Threat Modeling (MITRE ATT&CK), Penetration Testing (Hardware and Software), Incident Response

Scripting: Python, JavaScript, Bash, LaTeX

Languages: English (IELTS 8.0), Bengali (Native), Turkish (A1), German (A1, ongoing studies)